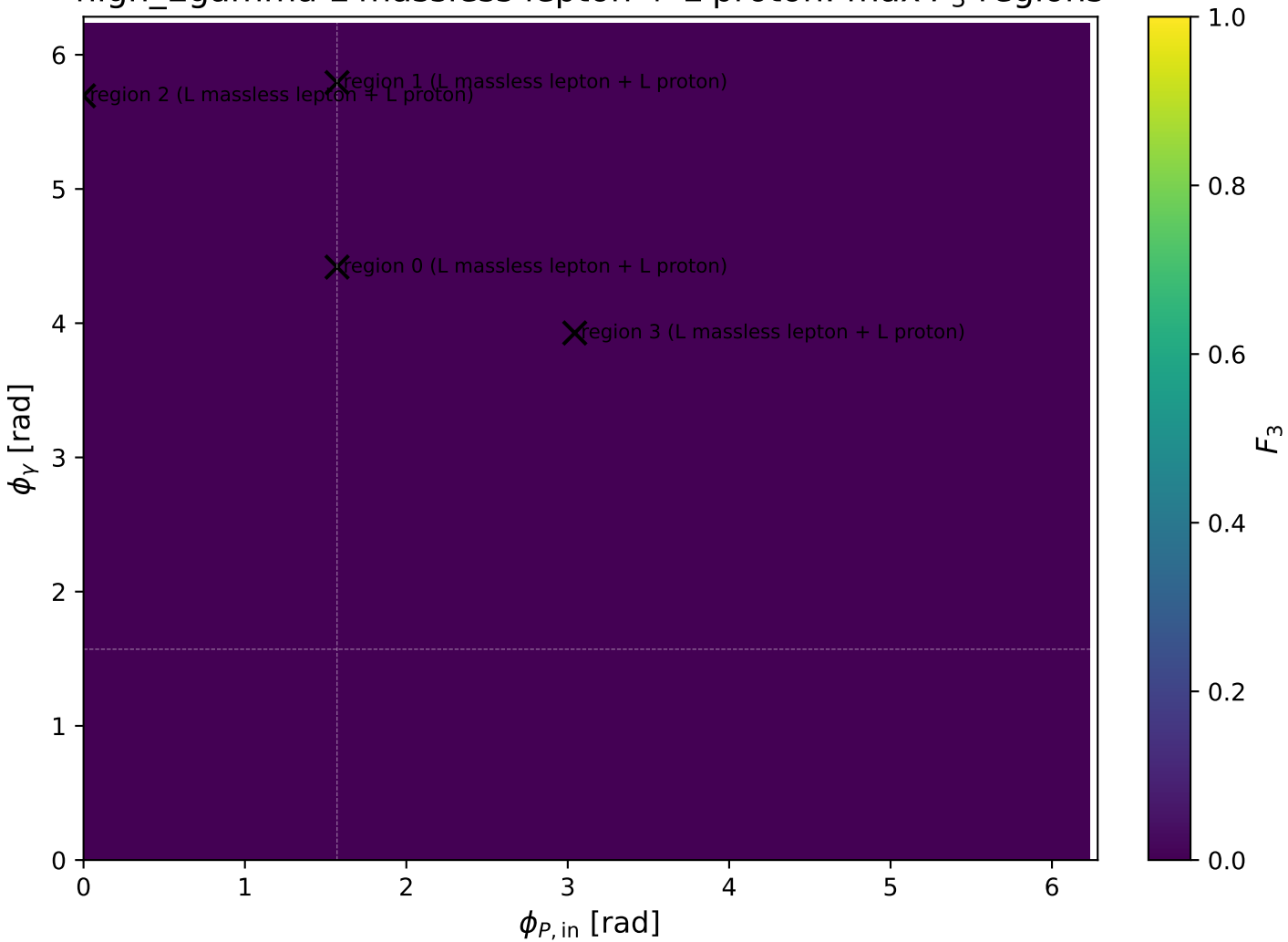
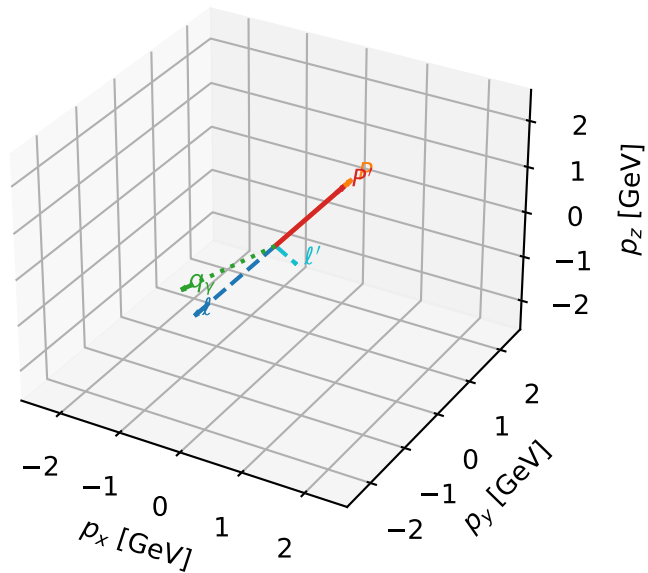


high_Egamma L massless lepton + L proton: max F_3 regions



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

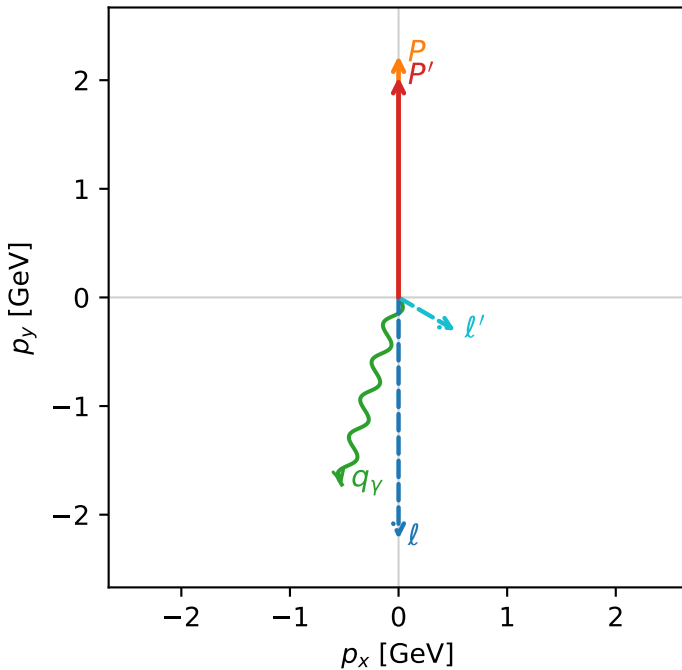


f3_high_egamma_ll_region_0 F_3=6.88031e-09
 region: high_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.02717$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_p=1.5708$
 $E_Y=1.8$, $\phi_Y=4.41786$
 $Q^2=1.33544$, $x_B=0.0816271$, $t=-0.00635022$, $W^2=15.9046$, $y=0.792799$
 $\theta(\ell', \gamma)=76.6286$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 0.6049$ $p_x= 0.5225$ $p_y= -0.3047$ $p_z= 0.0000$
 P' $E= 2.2337$ $p_x= 0.0000$ $p_y= 2.0272$ $p_z= 0.0000$
 q_Y $E= 1.8000$ $p_x= -0.5225$ $p_y= -1.7225$ $p_z= 0.0000$

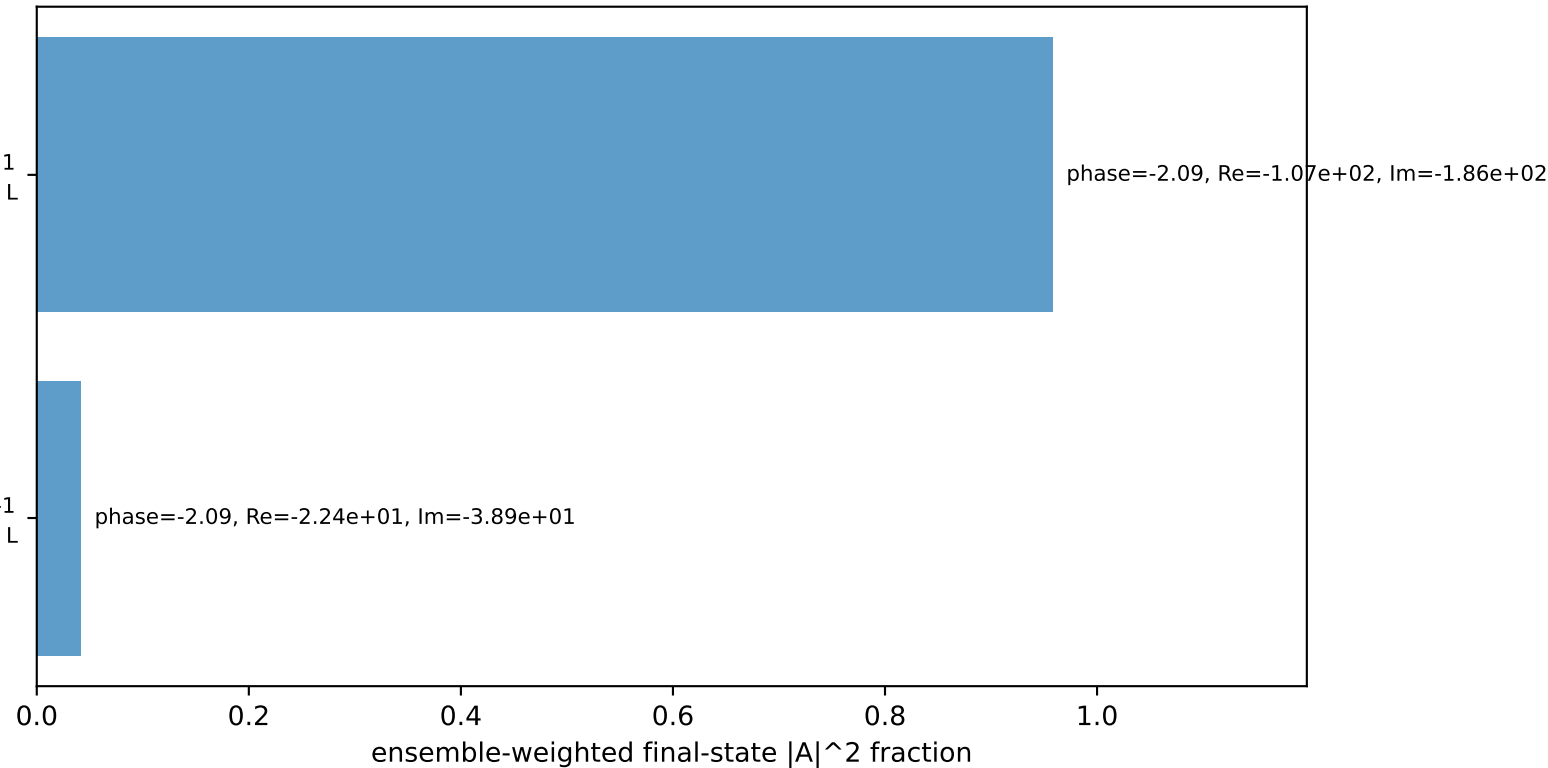
Transverse plane



$h_e=+1, h_p=-1, h_Y=+1$
 massless lepton L, proton L

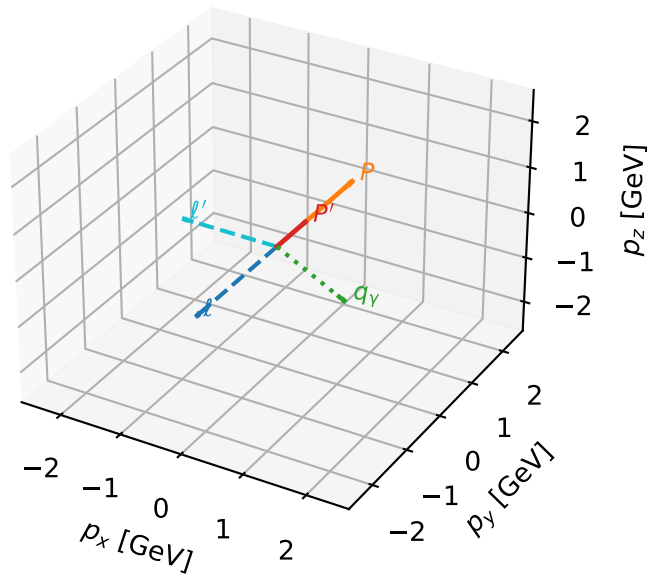
$h_e=+1, h_p=+1, h_Y=-1$
 massless lepton L, proton L

Leading initial-ensemble/final-state components (N=2, retained=100.0%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

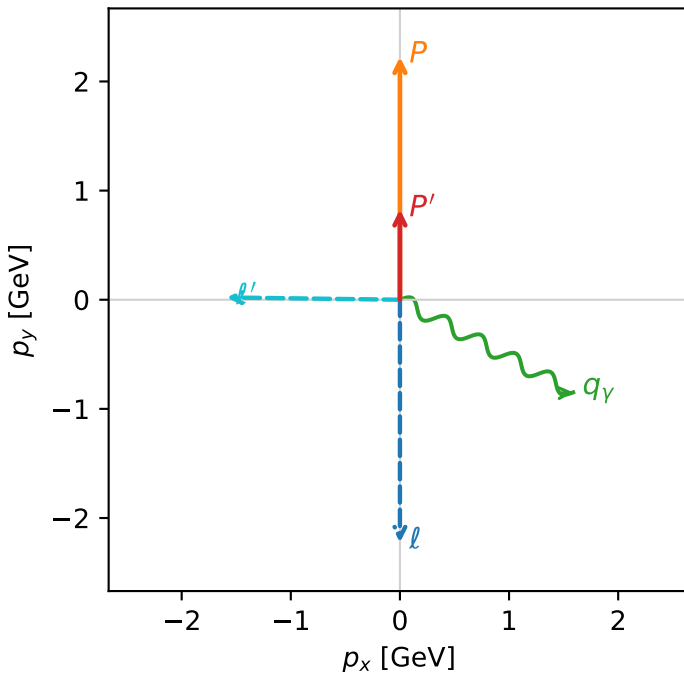


f3_high_egamma_ll_region_1 F_3=4.15877e-09
 region: high_Egamma_LL_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=0.827624$
 $\theta_{in}=1.57079$, $\phi_{\ell}=4.71239$, $\phi_P=1.5708$
 $E_Y=1.8$, $\phi_Y=5.79231$
 $Q^2=7.15589$, $x_B=0.547769$, $t=-0.59805$, $W^2=6.78766$, $y=0.633054$
 $\theta(\ell', \gamma)=152.629$ deg

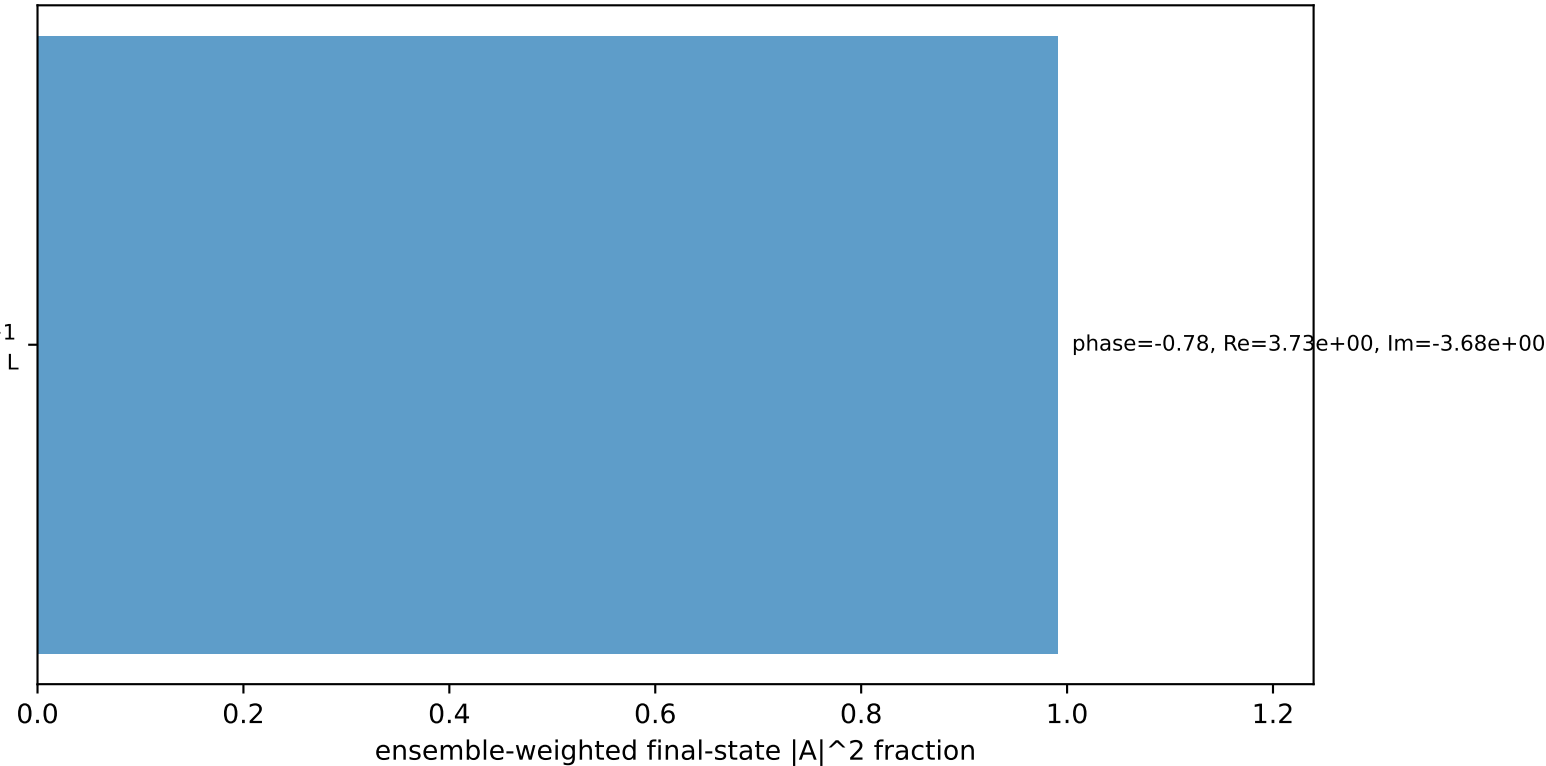
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.5876$ $p_x= -1.5875$ $p_y= 0.0209$ $p_z= 0.0000$
 P' $E= 1.2509$ $p_x= 0.0000$ $p_y= 0.8276$ $p_z= 0.0000$
 q_Y $E= 1.8000$ $p_x= 1.5875$ $p_y= -0.8485$ $p_z= 0.0000$

Transverse plane



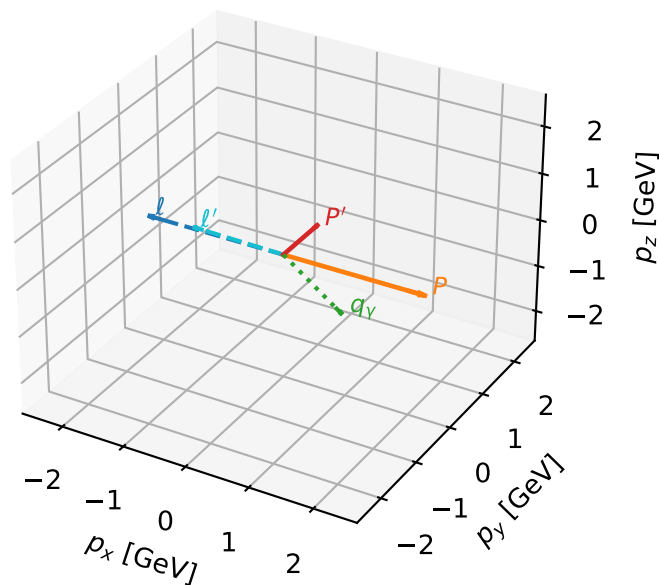
$h_e=+1, h_p=-1, h_Y=+1$
 massless lepton L, proton L

Leading initial-ensemble/final-state components (N=1, retained=99.2%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

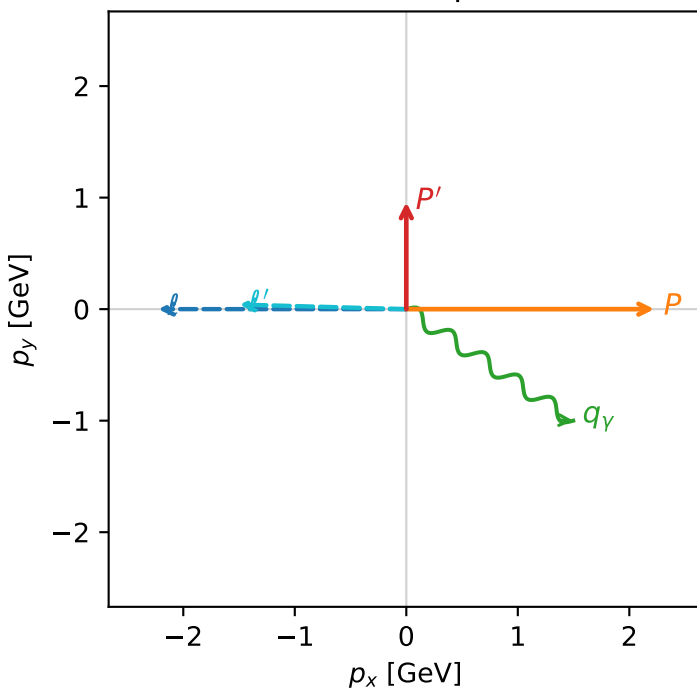


f3_high_egamma_ll_region_2 $F_3=3.82791e-09$
 region: high_Egamma_LL_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_{t_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=0.958776$
 $\theta_{in}=1.57079$, $\phi_l=3.14159$, $\phi_p=0$
 $E_\gamma=1.8$, $\phi_\gamma=5.69414$
 $Q^2=0.00252853$, $x_B=0.000374662$, $t=-4.71639$, $W^2=7.62614$, $y=0.327041$
 $\theta(l', \gamma)=147.829$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= -2.2244$ $p_y= -0.0000$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 2.2244$ $p_y= 0.0000$ $p_z= 0.0000$
 l' $E= 1.4972$ $p_x= -1.4966$ $p_y= 0.0413$ $p_z= 0.0000$
 P' $E= 1.3413$ $p_x= 0.0000$ $p_y= 0.9588$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= 1.4966$ $p_y= -1.0000$ $p_z= 0.0000$

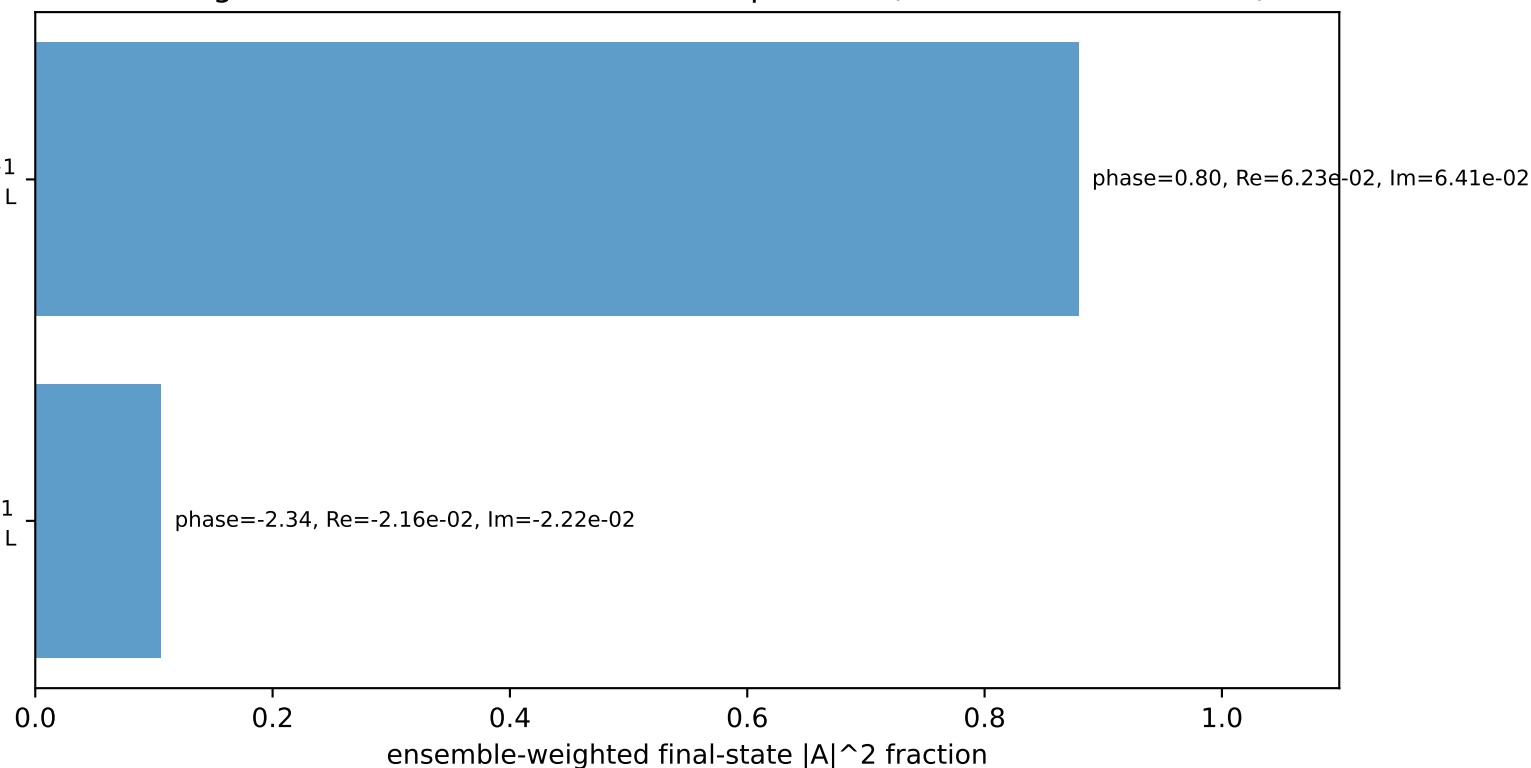
Transverse plane



$h_e=+1$, $h_p=+1$, $h_\gamma=+1$
 massless lepton L, proton L

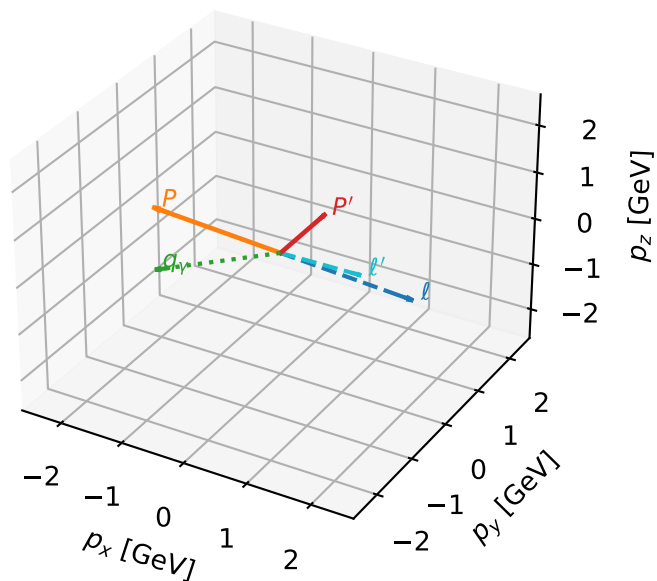
$h_e=+1$, $h_p=-1$, $h_\gamma=+1$
 massless lepton L, proton L

Leading initial-ensemble/final-state components (N=2, retained=98.5%)



L massless lepton + L proton characteristic max F_3 configuration

3D momenta

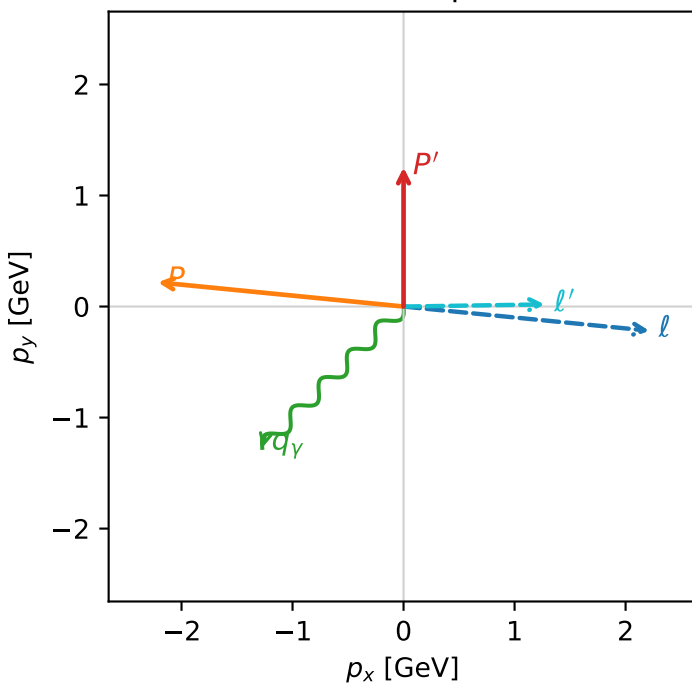


f3_high_egamma_ll_region_3 $F_3=3.39267\text{e-}09$
 region: high_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.25347$
 $\theta_{in}=1.57079$, $\phi_l=6.18501$, $\phi_p=3.04342$
 $E_\gamma=1.8$, $\phi_\gamma=3.92699$
 $Q^2=0.0363433$, $x_B=0.00410045$, $t=-5.25265$, $W^2=9.70675$, $y=0.429504$
 $\theta(l', \gamma)=135.87$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= 2.2137$ $p_y= -0.2180$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -2.2137$ $p_y= 0.2180$ $p_z= 0.0000$
 l' $E= 1.2729$ $p_x= 1.2728$ $p_y= 0.0193$ $p_z= 0.0000$
 P' $E= 1.5656$ $p_x= 0.0000$ $p_y= 1.2535$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -1.2728$ $p_y= -1.2728$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=97.9%)

